A local non-derivation forces a non-inner derivation

Literature-implied answer to Remark 3.9 of arXiv:0901.2983

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Status. Full affirmative answer implied by 2013–2014 literature. The deduction is recorded here for clarity and is not claimed as a new theorem.

Source question

For a von Neumann algebra $\mathcal{M}$ with faithful normal semifinite trace τ, Albeverio–Ayupov–Kudaybergenov–Nurjanov ask whether, for either

$$\mathcal{A} = S(\mathcal{M}) \quad \text{or} \quad \mathcal{A} = S(\mathcal{M}, \tau),$$

the existence of a local derivation which is not a derivation implies the existence of a non-inner derivation.

derivations of the form D_δ (see Section 2).

The implication (iv) $\Rightarrow$ (i) is true and follows from Proposition 2.7.

Finally, the implication (iv) $\Rightarrow$ (ii) is an open problem in the general case.

Later results

Two post-question theorems eliminate all non-derivational local maps on a von Neumann algebra without abelian direct summands.

- Hadwin–Li–Li–Ma, Theorem 1(1), applies to $S(\mathcal{M})$: a ring containing $\mathcal{M}$ whose nonabelian type summands form a separating central family has only derivational local derivations.
- Mukhamedov–Kudaybergenov, Theorem 3.1, applies to $S(\mathcal{M}, \tau)$: this is a solid $*$-subalgebra containing every finite-trace projection.

and a von Neumann algebra $\mathcal{M}_\infty$ which is the direct sum of algebras of type I_∞, II, and III (not all summands need be present). Call the corresponding central projections P_n with $1 \leq n \leq \infty$.

THEOREM 1. *Suppose $\mathcal{M}$ is a von Neumann algebra on a Hilbert space H. Then*

(1) *If $\mathcal{M}_1 = 0$ and $\mathcal{R}$ is a ring containing $\mathcal{M}$ with the same identity as $\mathcal{M}$ such that $\mathcal{P} = \{P_n : 2 \leq n \leq \infty\}$ is a separating family of central idempotents for $\mathcal{R}$, then every local derivation on $\mathcal{R}$ is a derivation.*

(2) *If $\mathcal{M}_1$ is discrete, then every local derivation on the algebra of closed densely defined operators affiliated with $\mathcal{M}$ is a derivation.*

The main result of this section is the following

Theorem 3.1. *Let M be a semi-finite von Neumann algebra without abelian direct summands and let τ be a faithful normal semi-finite trace on M. Suppose that $\mathcal{A}$ is a solid $*$-subalgebra in $S(M, \tau)$ such that $p \in \mathcal{A}$ for all $p \in P_\tau(M)$. Then every local derivation Δ on the algebra $\mathcal{A}$ is a derivation.*

For the proof of this theorem we need several lemmata

Affirmative deduction

Theorem 1. *Let $\mathcal{M}$ be a von Neumann algebra with a faithful normal semifinite trace τ, and let $\mathcal{A}$ denote either $S(\mathcal{M})$ or $S(\mathcal{M}, \tau)$. If $\mathcal{A}$ admits a local derivation which is not a derivation, then $\mathcal{A}$ admits a non-inner derivation.*

Proof. Let z be the maximal central projection such that $z\mathcal{M}$ is abelian, and put $\mathcal{N} = (1 - z)\mathcal{M}$. Then $\mathcal{N}$ has no abelian direct summand and

$$\mathcal{A} = z\mathcal{A} \oplus (1 - z)\mathcal{A}. \tag{1}$$

First recall the elementary central-homogeneity fact. If e is a central projection and Δ is a local derivation on $\mathcal{A}$, then

$$\Delta(ex) = e\Delta(x). \tag{2}$$

Indeed, choose a derivation D with $\Delta(ex) = D(ex)$. Every derivation annihilates a central idempotent, so $\Delta(ex) = eD(x) \in e\mathcal{A}$. Applying the same argument to $(1 - e)x$ and using linearity gives (2).

Consequently any local derivation Δ decomposes as

$$\Delta = \Delta_z \oplus \Delta_{1-z}$$

along (1), and both restrictions are local derivations. If $\mathcal{A} = S(\mathcal{M})$, Hadwin–Li–Li–Ma's theorem makes Δ_{1-z} a derivation on $S(\mathcal{N})$. If $\mathcal{A} = S(\mathcal{M}, \tau)$, Mukhamedov–Kudaybergenov's theorem makes it a derivation on $S(\mathcal{N}, \tau|_{\mathcal{N}})$.

Now suppose Δ is not a derivation. Its Leibniz defect splits along (1), while the $(1-z)$ component has zero defect. Hence Δ_z is not a derivation. But $z\mathcal{M}$ is commutative, and Theorem 3.8 of the source paper says that the existence of such a local derivation on $z\mathcal{A}$ is equivalent to the existence of a non-inner derivation d_z there.

Extend d_z by zero:

$$d(x) = d_z(zx), \qquad x \in \mathcal{A}.$$

Centrality of z makes d a derivation on $\mathcal{A}$. It is not inner: every inner derivation restricts to zero on the commutative central ideal $z\mathcal{A}$, whereas $d|_{z\mathcal{A}} = d_z \neq 0$. This proves the implication for both choices of $\mathcal{A}$. □

Classification and scope

For $S(\mathcal{M})$ the decisive later input appeared in arXiv:1311.0030 (2013); for $S(\mathcal{M}, \tau)$ it appeared in arXiv:1410.1619 (2014). The final central-summand argument combines those results with the source's own commutative Theorem 3.8. Thus the appropriate classification is *literature-implied full answer*, not a new solution. The argument assumes exactly the source setting of a faithful normal semifinite trace.

References

[1] S. Albeverio, Sh. A. Ayupov, K. K. Kudaybergenov, and B. O. Nurjanov, *Local Derivations on Algebras of Measurable Operators*, arXiv:0901.2983 (2009), Theorem 3.8 and Remark 3.9.

[2] D. Hadwin, J. Li, Q. Li, and X. Ma, *Local derivations on rings containing a von Neumann algebra and a question of Kadison*, arXiv:1311.0030 (2013), Theorem 1.

[3] F. Mukhamedov and K. Kudaybergenov, *Local derivations on subalgebras of τ-measurable operators with respect to semi-finite von Neumann algebras*, arXiv:1410.1619 (2014), Theorem 3.1.